

▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 6 summarizes the current fairness state for upload 6. Accuracy changed by +0.794 and fairness score changed by +0.727. Strong fairness improvement with no measurable accuracy loss. The best balanced model is Pipeline_mitigated_smote with combined score 0.754.

ACCURACY

0.794

▲ +0.794

FAIRNESS SCORE

0.727

▲ +0.727

COMBINED SCORE

0.754

Pipeline_mitigated_smote

DISPARATE IMPACT

0.500

Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
compas.csv	compas_dt_baseline.pkl	comparison

Fairness Metrics

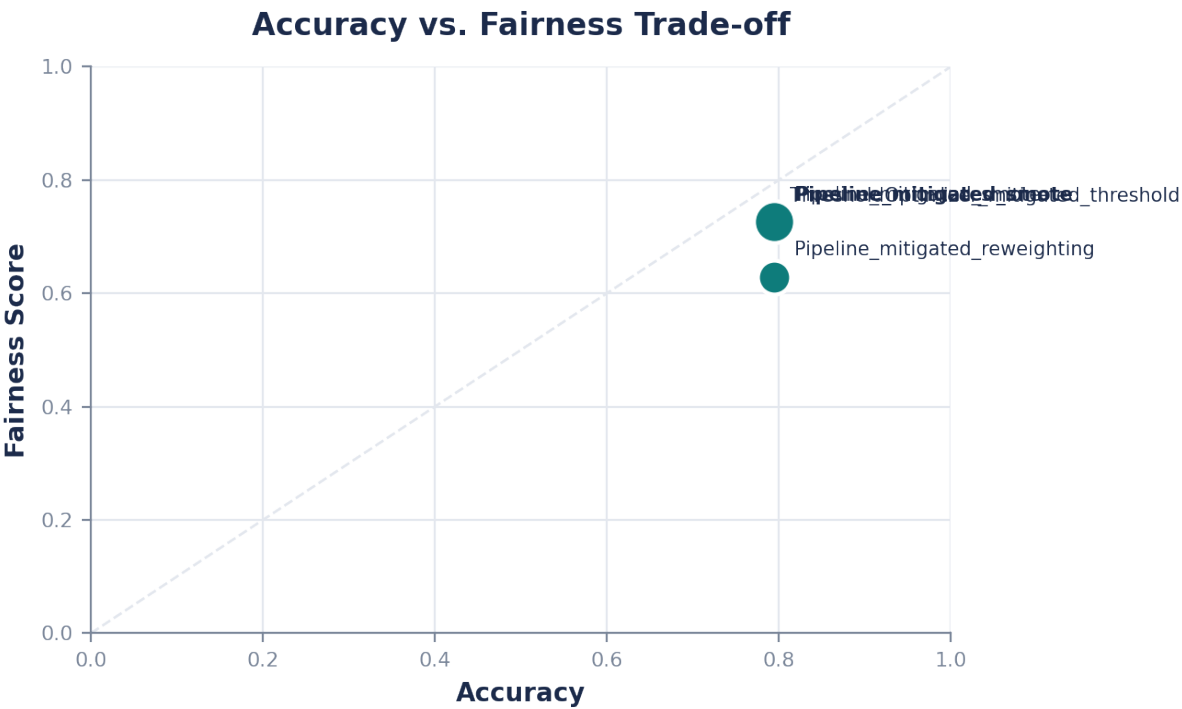
METRIC	VALUE	INTERPRETATION
Demographic Parity Difference (DPD)	0.250	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.296	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.500	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

Model Comparison

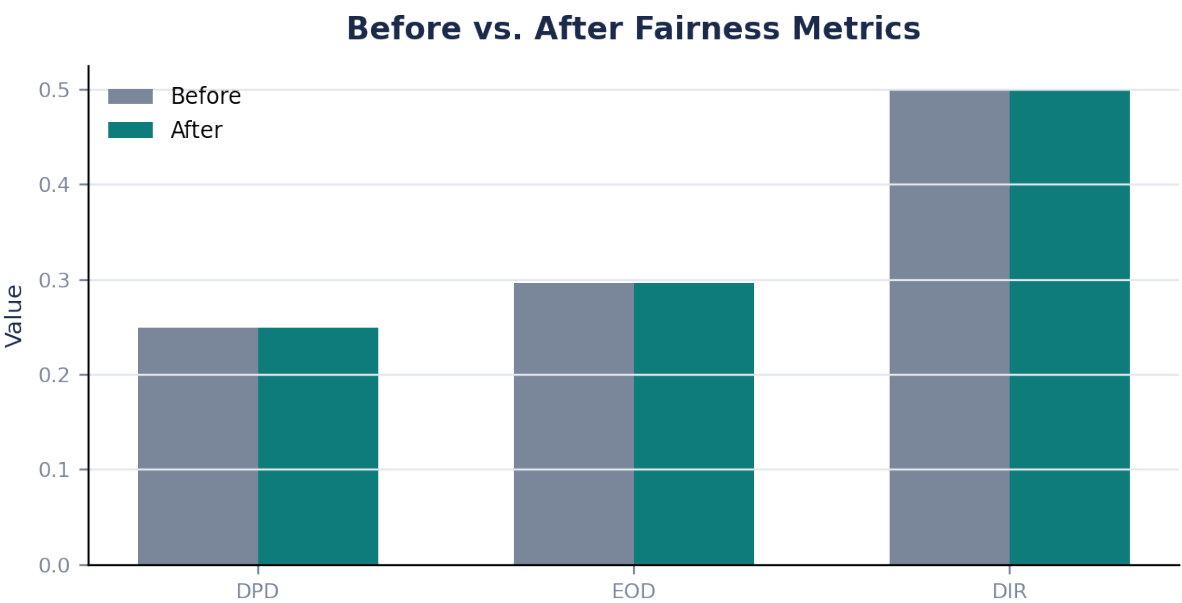
Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ Pipeline_mitigated_smote	0.794	0.727	0.250	0.296	0.500	0.754

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ Pipeline_mitigated_smote	0.794	0.727	0.250	0.296	0.500	0.754
ThresholdOptimizer_mitigated_threshold	0.789	0.724	0.296	0.256	0.297	0.750
Pipeline_mitigated_reweighting	0.794	0.629	0.337	0.405	0.394	0.695

Accuracy vs. Fairness Trade-off



Before vs. After Fairness Metrics



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.